CS & IT ENGINEERING

Discrete Mathematics

Mathematical Logic

DPP- 01

Discussion Notes

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[MCQ]

#Q. Which of the following is tautology?

A. $(\sim p \wedge (p \to q)) \to \sim q$

B. $\sim (p \to q)) \to \sim q$

C. $[(\sim p \wedge q) \wedge (q \to (p \to q)]] \to \sim r$

D. None of these

A) $(\sim p \wedge (p \to q)) \to \sim q$

$\equiv (\sim p \wedge (\sim p \vee q)) \to \sim q$

By absorption law $(p \vee (p \wedge q)) \equiv p$

$p \wedge (p \vee q) \equiv p)$

$\equiv \sim p \to \sim q$

B) $\sim (p \to q) \to \sim q$

$\equiv (p \wedge \sim q) \to \sim q$

$\equiv \sim p \vee q \vee \sim q$

$\equiv T$

$$((\sim p \wedge q) \wedge (q \rightarrow (p \rightarrow q))) \longrightarrow \sim r$$

$$\equiv ((\sim p \wedge q) \wedge (\underbrace{\sim q \vee \sim p \vee q}_{T})) \longrightarrow \sim r$$

$$\equiv (\sim p \wedge q) \longrightarrow \sim r$$

[MCQ]

#Q. The statement [PV (p ⟷ Q) V Q] is equivalent to

$\overset{F}{p} \vee \overset{T}{(\underset{F}{p} \leftrightarrow \underset{F}{q})} \vee \overset{F}{q} \equiv T$

$\equiv$

A. P

B. Q

C. A tautology

D. $(P \wedge Q)$

[MSQ]

#Q. Consider the following statement

✗ S_1: $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (r \Rightarrow p)$

S_2: $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$

A) S_1 is contingency

B) S_2 is tautology

C) S_1 and S_2 both contingency

D) S_1 and S_2 both Tautology

S_1 $((p \to q) \wedge (q \to r)) \to r \to p$

$\equiv ($

$\left.\begin{array}{l} p \to q \\ \wedge \; q \to r \end{array}\right\}$ Transitive

$p \to r \to r \to p$

$\not\equiv$

[MCQ]

#Q. Which of the following is valid?

$S_1: p \Rightarrow (q \vee r) \equiv (p \Rightarrow q) \vee (p \Rightarrow r)]$

$S_2: p \Rightarrow (q \wedge r) \equiv (p \Rightarrow q) \wedge (p \Rightarrow r)]$

A. S_1 is valid and S_2 is not valid

B. S_2 is not valid and S_2 is valid

C. Both S_1 and S_2 are valid

D. Neither S_1 and S_2 is valid

S_1: LHS:

$p \to (q \vee r) \equiv \underline{\sim p \vee q} \vee r$

RHS:

$(p \to q) \vee (p \to r)$

$\equiv \sim p \vee q \vee \sim p \vee r$

$= \sim p \underline{\vee q \vee r}$

S_2: LHS

$p \to q \wedge r = \sim p \underline{\vee (q \wedge r)}$

RHS:

$(p \to q) \wedge (p \to r)$

$= (\sim p \vee q) \wedge (\sim p \vee r)$

$= \sim p \vee (q \wedge r)$

[MSQ]

#Q. Which of the following is/are true.

A) $\{R \to S,\ P \to Q,\ R \cup P\} \Rightarrow S \cup Q$

B) $\{\sim R \to (S \to \sim T),\ \sim R \cup W,\ \sim P \to S,\ \sim W\} \Rightarrow T \to P$

C) $\{\sim P \cap Q,\ Q \to (P \to R)\} \Rightarrow \sim R$

D) $\{P \to R,\ P,\ Q \cup \sim R\} \Rightarrow Q$

A)
$$\left.\begin{array}{l} R \to S \\ P \to Q \\ R \vee P \end{array}\right\} \text{Constructive dilemma}$$
$$\overline{S \vee Q}$$

B)
$$\begin{array}{l} \overset{T}{\sim R} \to (S \to \sim \overset{T}{T}) \\ T\ \sim R \vee \overset{F}{W} \\ \sim P \to S \\ T\ \sim W \\ \hline T \to P \ \checkmark \end{array}$$

$S \to \sim T$
$\sim P \to S$

$\equiv \sim S \vee \sim T$
$P \vee S$

$$\left.\begin{array}{l} \equiv \sim S \vee \sim T \\ \underline{S} \vee P \end{array}\right\} \text{Resolution}$$

D) $\overset{T}{P} \rightarrow \overset{T}{R}$

$\overset{T}{P}$

$\overset{T}{Q} \vee \overset{F}{\sim R}$

$\sim Q$

$\sim T \vee P$

$\Rightarrow T \rightarrow P$

[MCQ]

#Q. Consider the following logical inferences.

I_1 : "I will study discrete math or I will study English literature"

"I will not study discrete math"

Inference : I will not study English literature

I_2 : "If A works hard then B or C will enjoy themselves"

"If B enjoys himself then A will not work hard"

"If C enjoys himself then D will not enjoy himself"

Inference: - If A work hard then D will not enjoy himself

A. Both I_1 and I_2 are correct inferences

B. I_1 is correct but I_2 is not a correct inference

C. I_1 is not correct but I_2 is a correct inference

D. Both I_1 and I_2 are not correct inferences

I_1 : p : I will study DM

Q : I will study Eng

1: $\overset{F}{p} \vee \overset{T}{q}$

2. $\sim p$ T

$\sim q$ ✓ q

I_2 :

A : A works hard

B : B enjoys himself

C : C ,, ,,

D : D ,,

1. $\overset{T}{A} \to (\overset{F}{B} \vee \overset{F}{C}) \longrightarrow F$

2. $\overset{F}{B} \to \sim\overset{F}{A}$ — T

3. $\overset{F}{C} \to \sim D \; F$ — T

$\underset{T}{A} \to \underset{F}{\sim D}$ F

$\}$ F

[MSQ]

#Q. Which of the following is/are logical equivalence?

I. $\sim(p \to q)$ — $p \wedge \sim q$

II. $\sim(p \to q) \wedge (q \to r)$

III. $p \wedge \sim \cancel{p}\, q$

IV. $(p \vee q) \to r$

$(p \wedge \sim q) \wedge (\sim q \vee r)$

$\equiv (p \wedge \sim q \wedge \sim q) \vee (p \wedge \sim q \wedge r)$

$\equiv \underline{(p \wedge \sim q)} \vee (p \wedge \underline{\sim q} \wedge r)$

$\equiv (p \wedge \sim q)$

A ~~I and II~~

B ~~I and III~~

C II and IV

D ~~II and III~~

$(p \vee q) \to r$

$\equiv (\sim p \wedge \sim q) \vee r$

[MCQ]

#Q. Consider the following statement

$S_1 : \sim(p \leftrightarrow q)$

$S_2 : p \leftrightarrow \sim q$

Which of the following is correct?

$$S_1 : \sim(p \Leftrightarrow q)$$

XOR

$$\equiv (\sim p \wedge q) \vee (p \wedge \sim q)$$

$$S_2 : p \Leftrightarrow \sim q$$

$$\equiv (p \wedge \sim q) \vee (\sim p \wedge q)$$

$$p \Leftrightarrow q$$

$$(p \wedge q) \vee (\sim p \wedge \sim q)$$

A. S_1 is tautology

B. S_2 is ~~concentration~~ contradiction

C. S_1 is equivalence to S_2

D. None of these

[MCQ]

#Q. Consider the following statement

$S_1 : \sim (p \vee (\sim p \wedge q))$

$S_2 : \sim p \wedge \sim q$

Which of the following is correct?

A S_1 is tautology

B S_2 is ~~concentration~~ contradiction

C S_1 is equivalence to S_2

D S_1 is not equivalence to S_2

$S_1 : \sim (p \vee (\sim p \wedge q))$

$\sim p \wedge \sim (\sim p \wedge q)$

$\sim p \wedge (p \vee \sim q)$

$\underbrace{(\sim p \wedge p)}_{F} \vee \underline{(\sim p \wedge \sim q)}$

$\equiv \sim p \wedge \sim q$

[MCQ]

#Q. $(p \to (q \to r)) \leftrightarrow ((p \to q) \to r)$ is

Associative

True — all are True

False — all are False

A Tautology

B Contingency

C Contradiction

D Can’s be determined

THANK - YOU